Scope, Related Work, and Limitations

Scope and epistemic boundary I

DuckRabbit is a typed, reproducible stimulus-construction platform. Its unit of work is an immutable request that yields a canonical image, audio buffer, video sequence, or audiovisual timeline, together with objective measurements and provenance. It is not an exhaustive ontology of all illusions, a perceptual theory, a participant database, or a substitute for a controlled psychophysics experiment. The live registry is therefore a deliberately bounded catalog: it enumerates the families for which this release has both a generator contract and a stated evidence boundary. Appendix A gives the complete current matrix; the absence of a family from that matrix is not a claim that the family is unimportant or unsupported in the wider literature.

Scope and epistemic boundary I

The central distinction is between a construction and an observation. A DuckRabbit generator can establish the dimensions, pixel values, sample values, frequency components, frame clock, declared synchronization offset, encoded hash, and decoded media facts of its output. It cannot, from those facts alone, establish what a person sees, hears, counts, localizes, groups, or reports. Every figure caption, evidence record, and claim-ledger entry preserves this boundary. In particular, a literature citation establishes a source-supported statement about a stimulus family or result under that source's conditions; it does not certify pixel- or task-identical replication by a DuckRabbit artifact.

Scope and epistemic boundary I

The historical foundation is intentionally worldwide and layered. The package does not present a single invention narrative: it places ancient Greek, medieval Arabic, classical Chinese, and early-modern European sources beside nineteenth-century psychophysics and modern experimental work. This is a scholarly orientation for separating physical construction, interpretation, and observer evidence; it is not a claim that the current source set exhausts the visual, auditory, or perceptual traditions of any region.

The modern label "visual illusion" sits on a longer experimental history than the package's contemporary review sources alone suggest. Oppel's mid-century catalogue of geometrical-optical illusions is now accessible with translation and commentary (Wade et al. 2017), and the primary nineteenth-century records include Zöllner's account of crossing-line distortions and Müller-Lyer's report of the arrow-wing configuration (Zöllner 1860; Müller-Lyer 1889). These sources are treated as historical anchors, not as evidence that a present-day raster is identical to an original plate or that its observer effect is invariant across conditions.

Visual families I

Gregory's classification is useful as a historical and conceptual orientation, but it is not a universally agreed ontology. DuckRabbit consequently stores visual modality, mechanism, perceptual signature, stimulus requirements, and evidence status as separate facets rather than treating a single label as an explanation (Gregory 1997). The visual atlas covers the implemented families currently registered in the package: ambiguous figure, contrast, apparent motion, Müller-Lyer, Poggendorff, Ponzo, Kanizsa-type subjective contour, Ebbinghaus contextual size, and Zöllner/Judd orientation-related geometry. "Coverage" here means one deterministic representative per live entry, not a claim that one raster captures the historical stimulus space.

The duck/rabbit construction is an explicit example of why that distinction matters. Brugger's historical analysis discusses variation among figure variants and observers; DuckRabbit therefore exposes a blend parameter and labels its output as a construction rather than promising a fixed alternation rate or universal bistability (Brugger 1999). The Müller-Lyer entry uses typed arrow geometry. Its engineering form is compatible with a family whose image statistics have been analyzed as potentially informative about image-source relationships, but the implementation does not adjudicate that account or infer a perceived-length report (Müller-Lyer 1889; Howe and Purves 2005).

The Poggendorff generator makes the occluding geometry and virtual-line orientation explicit. This is appropriate to a literature in which orientation estimation and filtering accounts are theoretically relevant, while avoiding the stronger claim that a particular raster reproduces a published bias without the same observers, viewing conditions, and response task (Morgan 1999). The Ponzo construction is similarly bounded: the converging context and test bars are deterministic, whereas the literature contains multiple explanations of Ponzo-like effects rather than one settled mechanism (Fisher 1967; Yildiz et al. 2022).

Visual families I

Kanizsa-style subjective contours are represented as an illusory-contour construction with explicit inducer geometry. The historical account motivates the family label, while modern reviews place contour completion within a wider literature on grouping, figure-ground organization, attention, and neural mechanisms (Kanizsa 1976; Wagemans et al. 2012). Neither source licenses a claim that the generated mask produces a uniform contour percept across observers. The Ebbinghaus entry controls target and surround geometry; classic work demonstrates that judged size depends on context, contour, and comparison conditions, while later work shows that motion and other parameters can alter the effect (Weintraub 1979; Mruczek et al. 2015). DuckRabbit's static default is therefore an engineering baseline rather than a task-identical replication. Zöllner/Judd geometry is likewise retained as a source-linked line arrangement. Zöllner's 1860 report supplies the historical primary anchor, while later work supplies a modern analysis of spatial filtering (Zöllner 1860; Earle and Maskell 1995). Its typed line lengths and orientations make the spatial stimulus auditable, while orientation judgments remain outside the package contract.

Finally, apparent motion is included in the visual coverage panel because its defining engineering object is temporal succession, not merely a static pattern. The package verifies frame order, frame rate, timestamps, and frame-to-frame differences. Wertheimer's foundational work and Sekuler's later analysis motivate the family distinction; neither source turns a generated frame strip into a universal report of motion (Wertheimer 1912; Sekuler 1996).

Auditory families I

The auditory catalog separates harmonic construction, spectral completion, pitch-class context, dichotic channel assignment, and continuity/masking. A Shepard-like signal is represented through additive partials and explicit envelopes, following a historical literature of tone psychology and later work on assimilation to an internalized musical scale (Stumpf 1883; Shepard and Jordan 1984). The canonical buffer exposes sample rate, amplitude bounds, channel count, partial frequencies, and envelope parameters. It does not determine a listener's perceived pitch height or direction.

The missing-fundamental construction removes a low component while retaining harmonically related partials. This makes the spectral condition reproducible; the associated pitch interpretation remains a listener-level question (Zatorre 2005). Tritone and octave entries preserve channel structure, phase, frequency relationships, and timing in typed parameters. Deutsch's foundational accounts and Repp's analysis of spectral-envelope and context effects motivate the evidence records, while also making listener and context dependence central limitations (Deutsch 1986, 1974; Repp 1997).

Auditory continuity is represented as an interrupted target with a typed masker and gap. Warren's classic perceptual-restoration result and Riecke and colleagues' separation of sensory and decisional contributions justify the family's inclusion, but a generated masker is not evidence that a listener will report an uninterrupted sound (Warren 1970; Riecke et al. 2011). The audio atlas therefore reports RMS, peak, spectral centroid, bandwidth, channel layout, and exact sample-level provenance—not continuity, pitch, or stream judgments.

Audiovisual and temporal binding families I

Audiovisual constructions require a shared clock and an explicit sign convention for offsets. The sound-induced-flash entry creates a typed visual event and one or more audio events; the evidence record links it to the primary demonstration and to a review of the broader literature (Shams et al. 2000; Hirst et al. 2020). The package can verify event timestamps, sample/frame clocks, and declared offsets. It does not infer a reported flash count, and it does not assume that the same temporal window applies across displays, headphones, latencies, or observers.

The spatial ventriloquist construction treats spatial discrepancy as a parameter and records the channel and display requirements. Reviews describe the ventriloquist illusion as a tool for studying multisensory processing, but the review's scope is not a license to claim spatial capture from a file alone (Bruns 2019). More general audiovisual accounts describe integration and segregation as a causal-inference problem shaped by temporal regularities and signal reliability (Noppeney and Lee 2018). That framework clarifies why a declared offset is an experimental input, not an observer-level outcome. Temporal ventriloquism receives a separate temporal-binding signature rather than being collapsed into spatial capture. Vroomen and de Gelder manipulated sound–flash timing in a flash-lag task and reported timing-dependent changes under those experimental conditions; Hartcher-O'Brien and Alais studied temporal ventriloquism in a purely temporal context (Vroomen and Gelder 2004; Hartcher-O'Brien and Alais 2011). DuckRabbit implements the stimulus-side timing contract and leaves the observer-side temporal-recalibration estimate to a future study.

McGurk remains `input_required`. The classic speech study motivates the family, but a lawful and reproducible implementation requires checksummed speech and video fixtures, licensing or consent records, a precise preprocessing contract, and an ethical validation protocol (McGurk and MacDonald 1976). Cataloguing the gap is more informative than silently substituting an unrelated synthetic voice or claiming that a generic audiovisual mismatch is a McGurk replication.

For each entry, the evidence matrix records a source role, exact source-supported claim, engineering basis, limitation, and audit status. The roles distinguish primary demonstration, review or synthesis, theoretical account, engineering basis, limitation, and input gap. This prevents three common category errors: treating a theory as settled mechanism, treating a review as validation of new code, and treating a generator's deterministic output as a participant result.

From literature to engineering contract I

The package's literature fidelity is therefore best described as "family-linked, parameterized engineering construction." Some defaults are historically motivated; none should be read as a claim of pixel identity unless that identity is separately established. The checked-in scholarship snapshot provides offline structural validation, including citation-key and DOI/URL format checks. The explicit network audit adds resolver observations and metadata matching; reachability alone is not treated as bibliographic verification. This is a reproducible evidence boundary, not a claim that every source is equally accessible or that theoretical disputes have been resolved.

Limitations and future observer work I

The package does not claim clinical validity, universal effect sizes, cross-cultural invariance, perceptual equivalence across displays or headphones, or observer-level truth from objective media metrics. Playback level, gamma and display calibration, viewing distance, refresh rate, stereo separation, room acoustics, audio transducer response, attention, expectation, language, expertise, and task can all matter. Codec behavior and device latency can introduce additional differences even when decoded media facts match within the declared tolerance.

The synthetic observer diagnostic is intentionally not a substitute for a real vision model or human data. It is a serialized, hand-specified feature function with `training_data=none`, `calibration = analytic`, and `human_data=false`; its metamorphic and sensitivity checks test orchestration, not visual consciousness or human discrimination. A future observer study must pre-register the response scale, reference condition, estimand, exclusions, missing-data rule, randomization seed, display and playback controls, and uncertainty procedure. It must also report the participant and item sampling frame rather than importing a model-output probability as an assumed effect size. The observer harness is consequently study-ready scaffolding, not a result set.

Limitations and future observer work I

No participant data are bundled with DuckRabbit. The data-availability boundary is intentional: canonical artifacts, encoded fixtures where lawful, source-data sidecars, and deterministic synthetic diagnostics are software outputs; observer outcomes require separately governed data collection. The package is authored by Daniel Ari Friedman and is archived under DOI `10.5281/zenodo.21419693`; release metadata and software-citation guidance are generated from the same identity contract as the manuscript.